# Type MS · Write a short number program from a list of requirements

## Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 1 of 6 · find part whole in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 | **Award 1 mark for each of:** | | | |
| | A1 | At least one variable with a meaningful name (1) | | |
| | A2 | At least one condition and price correct (ignore order) (1) | | |
| | A3 | All conditions and prices correct (1) | | |
| | A4 | Calculation for Total cost is correct (1) | | |
| | A5 | Price per textbook and total cost displayed (1) | | |
| | A6 | Calculation of cost and display of output are not included in the if statement (1) | | (6) |

Code examples

Python

```
# Q03

#  Initialise variables
quantity = 0
price = 0
cost = 0

#  Print prompt and take number of textbooks required
quantity = int(input("Please enter the quantity of books required: "))

#  Generate and display the price per textbook and the total cost
if quantity >= 1 and quantity <= 5:
    price = 20
elif quantity < 10:
    price = 15
else:
    price = 12

cost = price * quantity
print("The price per book is",price)
print("The total cost is",cost)
```

## Type MS · Write a short number program from a list of requirements

### Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 2 of 6 · find part whole in this mark scheme extract

| Question | mp | Answe | Additional Guidance | Mark |
|---|---|---|---|---|

# Type MS · Write a short number program from a list of requirements

## Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 3 of 6 · find part whole in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 6 | A1 | At least two appropriate subprograms used | | (1) |
| | A2 | Repeat guess until the word has been guessed **OR** no attempts left | | (1) |
| | A3 | Generate and display the number of attempts left | Must see working correctly | (1) |
| | A4 | Request input of the word | | (1) |
| | A5 | Validate the input of the word to ensure it is the same length as the random word | | (1) |
| | A6 | Check to see if the guess matches the random word | | (1) |
| | A7 | If the word has been guessed, display message including the random word and number of attempts | | (1) |
| | A8 | Keep track and display letters that are in the word | Must see working | (1) |
| | A9 | Keep track and display letters that are not in the word | Must see working | (1) |
| | A10 | Letters appear only once in each list | Must see working | (1) |
| | A11 | Display lose message including the random word | | (1) |

| Band 1 (1-3 marks) | Band 2 (4-6 marks) | Band 3 (7-9 marks) | Mark |
|---|---|---|---|
| Little attempt to decompose into component parts | Some attempt to decompose into component parts | The problem has been decomposed into component parts | |
| Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | (9) |

# Type MS · Write a short number program from a list of requirements

## Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 4 of 6 · find part whole in this mark scheme extract

Code examples

Python

Subprograms

```
# Add your subprograms here
def checkInput(pWordLength):
    # Validate input to make sure it is the same length as the secret word
    check = ""
    while len(check) != pWordLength:
        check = input("Enter your guess. The secret word has " +str(pWordLength)+ " letters: ")
    return(check)

def checkLettersInWord(pGuess, pWordToGuess, pLetters, type):
    for letter in pGuess: # Check each letter in the guess
        if type == 0: # The check to carry out is for correct letters
            # If the letter is in the secret word and not already in the correctletters, add it
            if letter in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
        else: # The check to carry out is for wrong letters
            # If the letter is not in the secret word and not already in the wrongLetters, add
            if letter not in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
    return pLetters

def display(pCorrectLetters, pWrongLetters):
    # Display the correct and wrong letters
    if len(pCorrectLetters) > 0:
        print("Letter(s) in the secret word:",pCorrectLetters)

    if len(pWrongLetters) > 0:
        print("Letter(s) not in the secret word:",pWrongLetters)
```

# Type MS · Write a short number program from a list of requirements

## Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 5 of 6 · find part whole in this mark scheme extract

**Main program**

```
# Add your main program code here
wordLength = len(wordToGuess)
maxAttempts = wordLength + 3
numAttempts = 0
correctLetters = ""
wrongLetters = ""
guessed = False

# Loop until the no attempts are left or the secret word has been guessed
while numAttempts < maxAttempts and guessed == False:
    print("You have",maxAttempts - numAttempts,"attempts to guess the secret word.")
    numAttempts = numAttempts + 1

    guess = checkInput(wordLength)  # Validate the input

    if guess == wordToGuess: # Check to see if the secret word has been guessed
        guessed = True
        print("Well done. You guessed the secret word was",wordToGuess,"in",numAttempts,"attempts.")

    else: # If it hasn't been guessed check for correct and wrong letters
        correctLetters = checkLettersInWord(guess,wordToGuess, correctLetters, 0)
        wrongLetters = checkLettersInWord(guess,wordToGuess, wrongLetters, 1)
        display(correctLetters, wrongLetters)

# Display game over message if attempts have run out and the secret word has not been guessed
if numAttempts == maxAttempts and guessed == False:
    print("Game over. You did not guess that the secret word was",wordToGuess)
```

## Type MS · Write a short number program from a list of requirements

### Answer 1 of 6 · Question 3 part whole · 6 marks

2022 JUN · Paper 02 · extract 6 of 6 · find part whole in this mark scheme extract

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## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 1 of 17 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| Q3(b) | More memory/storage required (1)<br>Because not sorted in place / not efficient / many steps / division of data set / is recursive (1) | | (2) |

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 2 of 17 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| Q3(a) | Award 1 mark each for:<br>1. File opened for input (1)<br>2. File opened for output (1)<br>3. Loop for processing data (1)<br>4. Check for '@' symbol (1)<br>5. Write invalid address to output file (1)<br>6. Close files (1) | Amended code should be saved as Q03aFINISHED.<br>Do not penalise candidates who attempt more than the stated requirements. | (6) |

## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 3 of 17 · find part c in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 4 of 17 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| Q3(c) | Award 1 mark for each section of code that:<br>1. Asks for input of a number and stores it in a variable with a meaningful name (1)<br>2. Checks number is between 1 and 50 (1)<br>3. Calculates square and cube of number (1)<br>4. Displays suitable labels e.g. 'Number', 'Square' and 'Cube' and relevant values (1)<br>For the whole code:<br>5. Compiles and runs without logical or syntax errors (1)<br>6. Efficient in terms of computation, storage and selection of programming constructs (1) | Amended code should be saved as Q03cFINISHED.<br><br>Allow follow through for mark point 4 when square and cube are incorrectly calculated.<br><br>Credit alternative coding which produces same results.<br><br>Accept use of exponent operators:<br>C# e.g. Math.Pow (2, 3) = 8<br>Java e.g. Math.pow (2, 3) = 8<br>Python e.g. 2**3 = 8. | (6) |

| Question | Designated marks: | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 5 of 17 · find part c in this mark scheme extract

**Java Example code**

# Type MS · Write a short number program from a list of requirements

## Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 6 of 17 · find part c in this mark scheme extract

3(a)

```
package q03aFINISHED;

import java.io.BufferedReader;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.PrintWriter;
import java.io.UnsupportedEncodingException;
import java.util.Scanner;

public class Q03aFINISHED
{
    public static void main(String[] args) throws FileNotFoundException,UnsupportedEncodingException
    {
        String email="";
        PrintWriter errorFile;

        //  Open the input file

        Scanner emailFile = new Scanner(new BufferedReader(new FileReader("Email.txt")));

        //  Open the output file

        errorFile = new PrintWriter("Error.txt", "UTF-8");

        //  Find errors and write to output file

        while (emailFile.hasNextLine())
        {
            email = emailFile.nextLine();
            if (!email.contains("@"))
            {
                errorFile.println(email);
            }
        }

        //  Close files

        emailFile.close();
        errorFile.close();
    }
}
```

# Type MS · Write a short number program from a list of requirements

## Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 7 of 17 · find part c in this mark scheme extract

| | | |
|---|---|---|
| 3(c) | Note: Line 19 is an alternative response to line 18<br><br>`package q03cFINISHED;`<br><br>`import java.util.Scanner;`<br><br>`public class Q03c`<br>`{`<br>`  public static void main(String[] args)`<br>`  {`<br>`  // Write your code here`<br>`  Scanner input = new Scanner(System.in);`<br>`  int myValue = 1;`<br>`  while (myValue >= 1 && myValue <= 50)`<br>`  {`<br>`  System.out.println("Enter a value from 1 to 50: ");`<br>`  myValue = input.nextInt();`<br>`  if (myValue >= 1 && myValue <= 50)`<br>`  {`<br>`  System.out.println("Value " + myValue + " Square " + myValue * myValue + " Cube " + myValue * myValue * myValue);`<br>`  System.out.println("Value " + myValue + " Square " + Math.pow(myValue, 2) + " Cube " + Math.pow(myValue, 3));`<br>`  }`<br>`  }`<br>`  }`<br>`}` | |

# Type MS · Write a short number program from a list of requirements

## Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 8 of 17 · find part c in this mark scheme extract

5

```
        int total = 0;
        double average;
        int books[]  = new int[libraryRecord.length];
        int gold = 0, silver = 0, bronze = 0;
        int goldID = 0, silverID = 0, bronzeID = 0;

        System.out.println("Pupil ID of pupils who have read less than 10 books");

            for (int i = 0; i < libraryRecord.length; i++)
            {
                books[i] = Integer.parseInt(libraryRecord[i][3]);
                if (books[i] > gold)
                {
                    bronze = silver; silver = gold; gold = books[i]; goldID = i;
                }
                else if (books[i] > silver)
                {
                    bronze = silver; silver = books[i]; silverID = i;
                }
                else if (books[i] > bronze)
                {
                    bronze = books[i]; bronzeID = i;
                }
                total = total + books[i];

                if (books[i] < 10)
                {
                    System.out.println(libraryRecord[i][0]);
                }
            }
            System.out.println();

            System.out.println("Total books read by pupils = " + total);

            System.out.println();

            average = total / libraryRecord.length;

            System.out.println("Average books read by pupils = " + average);

            System.out.println();

            System.out.println("Gold winner is " + libraryRecord[goldID][1] + " " + libraryRecord[goldID][2]);
            System.out.println("Silver winner is " + libraryRecord[silverID][1] + " " + libraryRecord[silverID][2]);
            System.out.println("Bronze winner is " + libraryRecord[bronzeID][1] + " " + libraryRecord[bronzeID][2]);
```

## Type MS · Write a short number program from a list of requirements

Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 9 of 17 · find part c in this mark scheme extract

Python Example code

1(c)

```
# Q01c

count = 0

maxValue = 5
constantValue = 7

while (count < maxValue):
    print(count, count + constantValue)
    count = count + 1
```

1(f)

```
    if (shopIncome > 5000) or (assistantSales >= 10):
        print ("Assistant bonus = ", assistantSalary * 0.1)

    elif (shopIncome >= 2000) and (assistantSales >= 5):
        print("Assistant bonus = ", assistantSalary * 0.05)
```

## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 10 of 17 · find part c in this mark scheme extract

2(a)

```
# Initialise variables
counter = 1
answer = randint(1,10)
guess = 0

# Print prompt and take guess from user
guess = int(input("Enter a number from 1 to 10: "))

# Create while loop to check answer
while guess!=answer:
    counter = counter + 1
    if(guess > answer):
        print (guess, " was too high. Try again.")
    else:
        print (guess, " was too low. Try again.")

    guess = int(input("Guess again: "))

# Report the correct answer to the user and indicate the number of guesses
print ("You guessed", guess, "in", counter, "guesses")
```

## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 11 of 17 · find part c in this mark scheme extract

3(a)

```
Q03aFINISHED

#   Open file and input data
with open('Email.txt','r') as inputFile:
    emailList = inputFile.readlines()

#   Open file for output
outputFile = open('Error.txt', 'a')

#   Find errors and write to error log
for address in emailList:
    if not "@" in address:
        outputFile.write(address)

#   Close files
outputFile.close()
inputFile.close()
```

## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 12 of 17 · find part c in this mark scheme extract

3(c)

```
1  # Q03c
2
3  # Write your code below this line
4  myValue = 1
5
6  while (myValue >= 1 and myValue <= 50):
7      myValue = int(input("Enter a value from 1 to 50: "))
8      if (myValue >= 1 and myValue <= 50):
9          print("Value: ", myValue, "Square: ",myValue*myValue, "Cube: ", myValue*myValue*
           myValue)
10
11         print("Value: ", myValue, "Square: ", myValue**2, "Cube: ", myValue**3)
```

Note: Line 11 is an alternative response to line 9.

# Type MS · Write a short number program from a list of requirements

## Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 13 of 17 · find part c in this mark scheme extract

5

```
# Write your code below this line

total = 0
gold = 0
silver = 0
bronze = 0
goldID = 0
silverID = 0
bronzeID = 0

print ("Pupil ID of pupils who have read less than 10 books ")

for row in range(len(libraryRecord)):
    books = int(libraryRecord[row][3])
    if (books > gold):
        bronze = silver
        silver = gold
        gold = books
        goldID = row
    elif (books > silver):
        bronze = silver
        silver = books
        silverID = row
    elif (books > bronze):
        bronze = books
        bronzeID = row

    total = total + books

    if (books < 10):
        print ("\t",libraryRecord[row][0])

print ("\n")
print ("Total number of books read by pupils: " , total)

print ("\n")
print ("Average number of books read by pupils: " , total / len(libraryRecord[0]))

print ("\n")
print ("Gold medal winner is: \t\t", libraryRecord[goldID][1], " ", libraryRecord[goldID][2])
print ("Silver medal winner is: \t", libraryRecord[silverID][1], " ", libraryRecord[silverID][2])
print ("Bronze medal winner is: \t", libraryRecord[bronzeID][1], " ", libraryRecord[bronzeID][2])
```

## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 14 of 17 · find part c in this mark scheme extract

## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 15 of 17 · find part c in this mark scheme extract

## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 16 of 17 · find part c in this mark scheme extract

## Type MS · Write a short number program from a list of requirements

### Answer 2 of 6 · Question 3 part c · 6 marks

2019 JUN · Paper 02 · extract 17 of 17 · find part c in this mark scheme extract

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# Type MS · Write a short number program from a list of requirements

## Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part b in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 2 (a) | A1 | 2.3.4 | Set variable base to 50 or heightChk to true | Logic of algorithm must be followed as set out.<br>Alternatives must address each point.<br>Do not penalise candidates who attempt more than the stated requirements.<br>Don't penalise spelling mistakes and alternative wording of the output. | (10) |
| | A2 | 2.2.2 | Create WHILE DO loop | | |
| | A3 | 2.4.1 | Request input of height | | |
| | A4 | 2.2.2 | if statement checks if height >= 1 or <= 100 | | |
| | A5 | 2.2.2 | if statement checks if height is >= 1 and <=100 | | |
| | A6 | 2.2.2 | heightCheck set to false if condition is met | | |
| | A7 | 2.5.1 | Set area using given formula (0.5 x base x height) | | |
| | A8 | 2.4.1 | Display one of base, height and area with appropriate label or displays two of base, height and area without labels | | |
| | A9 | 2.4.1 | Display all of base, height and area with appropriate labels | | |
| | A10 | 1.1.6 | Executing and producing correct output | | |
| Code example | | | | | |
| Python | | | | | |

```python
#   Q02aFINISHED

#   Initialise variables

base = 50

#   Print prompt and take input from user

    heightChk = TRUE
    while heightChk:
        height = int(input("Enter the height (between 1 and 100): "))
        if (height >= 1 and height <= 100):
            heightChk = FALSE

#   Calculate area and print out values

    area = 0.5 * base * height

    print ("Base length of triangle is : ", base)
    print ("Height of triangle is : ", height)
    print ("Area of triangle is : ", area)
```

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## Type MS · Write a short number program from a list of requirements

### Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part b in this mark scheme extract

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part b in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 2 (b) | B1 | 2.2.1 | Meaningful variable names used (1) | The code example shown is one way of responding to the task. Other methods should be credited accordingly. | (7) |
| | B2 | 2.4.1 | Meaningful prompts for input (1) | | |
| | B3 | 2.4.1 | Input of length and width as whole numbers (1) | | |
| | B4 | 2.5.1 | Perimeter calculated (= 2 x length + 2 x width) (1) | | |
| | B5 | 2.5.1 | Gap deducted from perimeter (1) | | |
| | B6 | 2.4.1 | Meaningful output (1) | | |
| | B7 | 2.1.1 | Code is fit for purpose. Must have mark points 1-6 and be easy to read (1) | | |

Code example

Python

```
1   #   Q02bFINISHED
2
3   #   Request input
4
5   length = int(input("Enter the length: "))
6   width = int(input("Enter the width: "))
7
8   #   Calculate number of panels needed
9
10  panels = 2 * length + 2 * width - 4
11
12  #   Print out number of panels needed
13
14  print ("Number of panels needed: " , panels)
```

## Type MS · Write a short number program from a list of requirements

### Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 4 of 7 · find part b in this mark scheme extract

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

# Type MS · Write a short number program from a list of requirements

## Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part b in this mark scheme extract

For Q6, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|
| 6 | A1 | 2.2.1 | Initialise variable for total sales by member of staff (1) | Sales for each member of staff<br><br>George Taylor 28466<br>Fehlix Chayne 35250<br>Sumren Bergen 30186<br>Samira Beckle 37430<br>Nellie Bogart 20448<br>Cheryl Grouth 34383<br>Danuta Graunt 26421<br>Jaiden Deckle 21507<br>Jimran Caliks 32415<br>Deynar Derran 28590<br>==========<br><br>Total staff sales: 295096<br><br>Highest sales by: Samira Beckle with 37430<br><br>Second highest sales by: Fehlix Chayne with 35250<br>>>> | (11) |
| | A2 | 2.5.1 | Add succeeding months' sales to total sales (1) | | |
| | A3 | 2.1.6 | Display total sales by member of staff (1) | | |
| | A4 | 2.2.2 | Loop repeats for each member of staff (1) | | |
| | A5 | 2.2.1 | Running team total sales initialised (1) | | |
| | A6 | 2.2.2 | Add succeeding individual staff sales to team total (1) | | |
| | A7 | 2.1.6 | Display final running total (1) | | |
| | A8 | 2.1.6 | Staff member with highest individual sales identified (1) | | |
| | A9 | 2.1.6 | Staff member with second highest individual sales identified (1) | | |
| | A10 | 2.1.6 | Information about staff with highest sales displayed (1) | | |
| | A11 | 2.1.6 | Information about staff with second highest sales displayed (1) | | |

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## Type MS · Write a short number program from a list of requirements

### Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 6 of 7 · find part b in this mark scheme extract

| Band 1 (1-3 marks) | Band 2 (4-6 marks) | Band 3 (7-9 marks) | Mark |
|---|---|---|---|
| Little attempt to decompose the problem into component parts | Some attempt to decompose the problem into component parts | The problem has been decomposed into component parts | |
| Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | (9) |

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## Type MS · Write a short number program from a list of requirements

### Answer 3 of 6 · Question 2 part b · 7 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part b in this mark scheme extract

Code example

Python

```
allStaffTotal = 0
highestSales = 0
secondSales = 0
high = 0
second = 0

print ("Sales for each member of staff","\n\n")
for staff in staffSales:
    staffTotal = 0
    sales = 3
    while (sales < 9):
        staffTotal = staffTotal + staff[sales]
        sales = sales + 1
    print (staff[1],staff[2],staffTotal)
    if (staffTotal > secondSales):
        if (staffTotal > highestSales):
            highestSales = staffTotal
            high = staff
        else:
            secondSales = staffTotal
            second = staff
    allStaffTotal = allStaffTotal + staffTotal
print ("==========================\n\nTotal sales for all staff: ", allStaffTotal)

print ("\n\nHighest sales by: ",high[1],high[2], " with ", highestSales)
print ("\n\nSecond highest sales by: ",second[1],second[2], " with ", secondSales)
```

## Type MS · Write a short number program from a list of requirements

### Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 1 of 9 · find part e in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (a) | A1 | **Award 1 mark for any of:**<br>• Comments (1)<br>• Indentation (1)<br>• Meaningful variable/constant/subprogram names (1)<br>• White space (1) | | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 2 of 9 · find part e in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 1 (b) | B1<br>B2<br>B3 | **Award** 1 mark for each column:<br><br>**Error description** / **Logic** / **Syntax** / **Runtime**<br>Divide by 0 / / / ✓<br>Use x instead of * multiply / / ✓ /<br>Subtract 10 from 2 instead of 2 from 10 / ✓ / / | | (3) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 3 of 9 · find part e in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (c) | C1 | **The only correct answer is D**<br><br>*A is not correct because as it is a data structure*<br><br>*B is not correct because as there is no arrow looping back to the condition*<br><br>*C is not correct because as the flowchart does not include an operator* | | (1) |

## Type MS · Write a short number program from a list of requirements

### Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 4 of 9 · find part e in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

# Type MS · Write a short number program from a list of requirements

## Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 5 of 9 · find part e in this mark scheme extract

| | | | |
|---|---|---|---|
| 1 (d) | **Award 1 mark for each of:**<br>D1 = changed to == (1)<br>D2 Condition changed to less than / final two print statements swapped (1)<br>D3 else changed to else (1) | | (3) |

Code examples

C#

```
if (number1 == number2)
{
    Console.WriteLine("The numbers are equal");
}
else if (number1 < number2)
{
    Console.WriteLine("The highest number is " + number2 + " and the lowest number is " + number1);
}
else
{
```

Java

```
if (number1 == number2)
{
    System.out.print("Numbers are equal");
}
else if (number1 < number2)
{
    System.out.print("The highest number is " + number2 + " and the lowest number is " +number1);
}
else
```

Python

```
if number1 == number2:
    print("Numbers are equal")
elif number1 < number2:
    print("The highest number is", number2, "and the lowest number is", number1)
else:
```

## Type MS · Write a short number program from a list of requirements

### Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 6 of 9 · find part e in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 7 of 9 · find part e in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (e) | **Award 3 marks for:** | | | |
| | E1 | Prompt for a number to be input (1) | | |
| | E2 | Input stored in an integer variable (1) | | |
| | E3 | Loop used (1) | | |
| | E4 | Loop iterates only for the values 1 – 12 (1) | | |
| | E5 | Display is correct when program executes (1) | | (5) |

Code examples

C#

```
// Print prompt and get number from user
Console.Write("Enter the number: ");
int num = Convert.ToInt32(Console.ReadLine());

// Create loop to display the table
for (int count = 1; count <= 12; count++)
{
    Console.WriteLine(count + " x " + num + " is " + count * num);
}
Console.ReadLine();
```

Java

```
// Print prompt and get number from user
Scanner input = new Scanner(System.in);
System.out.print("Enter the number ");
int num = input.nextInt();

// Create loop to display the table
for (int count = 1; count <=12; count++)
{
    System.out.println(count + " x " + num + " is " + count * num);
}
```

## Type MS · Write a short number program from a list of requirements

### Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 8 of 9 · find part e in this mark scheme extract

| | |
|---|---|
| Python | `# Print prompt and get number from user`<br>`num = int(input("Enter the number: "))`<br><br>`# Create loop to display the table`<br>`for count in range(1, 13):`<br>`    print(count, "x" , num , "is" , count * num)` |

## Type MS · Write a short number program from a list of requirements

### Answer 4 of 6 · Question 1 part e · 5 marks

2021 NOV · Paper 02 · extract 9 of 9 · find part e in this mark scheme extract

| Question | mp | | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 1 of 11 · find part c in this mark scheme extract

| Question | Mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (a) (i) | A1 | C# and Python monthNumber/monthsAndDays/count/found (1)<br>Java monthNumber/monthsAndDays/count/found/input | | (1) |

## Type MS · Write a short number program from a list of requirements

### Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 2 of 11 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (ii) | A2 | monthsAndDays (1) | (1) |

## Type MS · Write a short number program from a list of requirements

### Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 3 of 11 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iii) | A3 | C#<br>• WriteLine, ToString, ToInt32, ReadLine<br>Java<br>• Print, ParseInt, nextLine, ToString, Equals<br>Python<br>• print/len/int/input (1) | (1) |

## Type MS · Write a short number program from a list of requirements

### Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 4 of 11 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iv) | A4 | Display/showMonthNameAndDays (1) | (1) |

## Type MS · Write a short number program from a list of requirements

### Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 5 of 11 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (v) | A5 | pMonth (1) | (1) |

Type MS · Write a short number program from a list of requirements

Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 6 of 11 · find part c in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) (vi) | A6<br>A7 | **Award 1 mark for each of:**<br><br>**Type of test data** / **Test data**<br>Boundary (1) / 1<br>Erroneous (1) / 15 | | (2) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 7 of 11 · find part c in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (b) | B1<br>B2 | **Award up to 2 marks for a linked explanation such as:**<br><br>• A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1)<br>• (Using parameters) enables reusable solutions (1) using different data values/arguments (1)<br>• Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1) | | (2) |

## Type MS · Write a short number program from a list of requirements

### Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 8 of 11 · find part c in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

# Type MS · Write a short number program from a list of requirements

## Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 9 of 11 · find part c in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) | **Award 1 mark for each of:** | | | |
| | C1 | Condition to check whether the year exactly divides by 4 or equivalent condition (1) | | |
| | C2 | Condition to check whether year exactly divides by 400 or equivalent condition (1) | | |
| | C3 | Condition to check whether year does not exactly divide by 100 or equivalent condition (1) | | |
| | C4 | Correct message for at least one of the conditions (1) | | |
| | C5 | Program executes and produces the correct output (1) | | |
| | C6 | Efficient solution (1) | | |
| | | | | (6) |
| Code examples | | | | |
| C# |  | | | |
| Java |  | | | |

```
if (year % 400 == 0)
{
    Console.WriteLine(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    Console.WriteLine(year + " is a leap year");
}
else
{
    Console.WriteLine(year + " is not a leap year");
}
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

## Type MS · Write a short number program from a list of requirements

### Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 10 of 11 · find part c in this mark scheme extract

| | |
|---|---|
| Python | `if year % 400 == 0:`<br>`  print(year,"is a leap year")`<br>`elif year % 4 == 0 and year % 100 !=0:`<br>`  print(year,"is a lear year")`<br>`else:`<br>`  print(year,"is not a leap year")` |

## Type MS · Write a short number program from a list of requirements

### Answer 5 of 6 · Question 3 part c · 6 marks

2021 NOV · Paper 02 · extract 11 of 11 · find part c in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 1 of 13 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3(a) | A1<br>A2<br>A3<br>A4 | Award **one** mark for correct column (A OR B)<br>Award **one** mark for correct column (NOT B)<br>Award **one** mark for first set of two cells in third column<br>Award **one** mark for second set of two cells in third column | Award follow through on last column only<br>5th column can be judged<br>• On columns 1 and 2 for MP3 and MP4 or<br>• as FT from columns 3 and 4 for MP3 and MP4<br>• award highest mark | 4 |

| A | B | A OR B | NOT B | (A OR B) AND (NOT B) |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 |

## Type MS · Write a short number program from a list of requirements

### Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 2 of 13 · find part b in this mark scheme extract

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 3 of 13 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3(b) | | Award **one** mark for each of: | | |
| | B1 | Correct output taken from array (1) | Must be from array i.e. Indexing used | |
| | B2 | Use of iteration to loop over every item in the array (1) | Award foreach loops<br>Accept any array | |
| | B3 | Use of appropriate method to step accurately, i.e. every other item starting at position 0 (1) | Any appropriate step method<br>Not hard coded index | **3** |

## Type MS · Write a short number program from a list of requirements

### Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 4 of 13 · find part b in this mark scheme extract

**C#**

```
// Q03bFINISHED

String[] starNames = {"Alasia", "Beid", "Castor", "Denebola",
                      "Electra", "Fafnir", "Gudja", "Haedus",
                        "Izar", "Jishui", "Kang", "Lich",
                        "Maia", "Nahn", "Ogma", "Peacock"};


// Write your code here
int start = 0;
int stop = starNames.Length;
int step = 2;

for (int index = start; index < stop; index = index + step)
{
    Console.WriteLine (starNames[index]);
}

// Alternative
for (int i=0; i<starNames.Length; i=i+2)
{
    Console.WriteLine (starNames[i]);
}

```

# Type MS · Write a short number program from a list of requirements

## Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 5 of 13 · find part b in this mark scheme extract

**Java**

```java
// Q03bFINISHED


public class Q03bFINISHED
{
    public static void main (String[] args)
    {
        String starNames[] = {"Alasia", "Beid", "Castor", "Denebola",
                              "Electra", "Fafnir", "Gudja", "Haedus",
                              "Izar", "Jishui", "Kang", "Lich",
                              "Maia", "Nahn", "Ogma", "Peacock"};

        // Write your code here
        int start = 0;
        int stop = starNames.length;
        int step = 2;

        for (int index = start; index < stop; index = index + step)
        {
            System.out.println (starNames[index]);
        }

        // Alternative
        for (int i=0; i<starNames.length; i=i+2)
        {
            System.out.println (starNames[i]);
        }

    }
}
```

# Type MS · Write a short number program from a list of requirements

## Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 6 of 13 · find part b in this mark scheme extract

**Python**

```python
# Q03bFINISHED

starNames = ["Alasia", "Beid", "Castor", "Denebola",
             "Electra", "Fafnir", "Gudja", "Haedus",
             "Izar", "Jishui", "Kang", "Lich",
             "Maia", "Nahn", "Ogma", "Peacock"]

# Write your code here

start = 0
stop = len (starNames)
step = 2

for index in range (start, stop, step):
    print (starNames[index])

# Alternative
for index in range (0, len (starNames), 2):
    print (starNames[index])
```

## Type MS · Write a short number program from a list of requirements

### Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 7 of 13 · find part b in this mark scheme extract

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|

## Type MS · Write a short number program from a list of requirements

### Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 8 of 13 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3(c) | | Award **one** mark for each: | | |
| | C1 | Take a string input (1) | • `aString = Console.ReadLine ();`<br>• `aString = myKeyboard.nextLine();`<br>• `aString = input ("Enter a number: ")` | **6** |
| | C2 | Complete the selection with 'if' (1) | | |
| | C3 | Convert string to integer (1) | • `aNum = Convert.ToInt32 (aString);`<br>• `aNum = Integer.parseInt (aString);`<br>• `aNum = int (aString)` | |
| | C4 | Any one correct set of relational operators (1)<br>ignore logical operator<br>first two tests<br>ignore test dealing with >= 1 | Python<br>• `(aNum <= 20) or (aNum >= 60)`<br>• `(aNum < 21) or (aNum > 59)`<br>• `(aNum > 30) and (aNum < 40)`<br>• `(aNum >= 31) and (aNum <= 39)` | |
| | C5 | Both logical operators used correctly in each test (1)<br>Ignore relational operators<br>First test: **or** (must attempt to address 20 and 60 with relational, even if C4 not awarded<br>Second test: **and** | C# and Java<br>• `((aNum <= 20) \|\| (aNum >= 60))`<br>• `(aNum < 21) \|\| (aNum > 59))`<br>• `((aNum > 30) && (aNum < 40))`<br>• `((aNum >= 31) && (aNum <= 39))` | |
| | C6 | Complete the third test (1) | All Languages:<br>`(aNum == 30)` | |

# Type MS · Write a short number program from a list of requirements

## Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 9 of 13 · find part b in this mark scheme extract

**C#**

```
// Q03cFINISHED

String aString = "";
int aNum = 0;

Console.WriteLine ("Enter a number:");

// Complete the line to take a string input
aString = Console.ReadLine ();

// Complete the selection statement
if (aString != "")
{
    // Convert the string to an integer
    aNum = Convert.ToInt32 (aString);
    if (aNum > 0)
    {
        // Complete the test
        if ((aNum <= 20) || (aNum >= 60))
        {
            Console.WriteLine ("Acceptable");
        }
        // Complete the test
        else if ((aNum >30) && (aNum < 40))
        {
            Console.WriteLine ("Centre");
        }
        // Complete the test
        else if (aNum == 30)
        {
            Console.WriteLine ("Perfect");
        }
        else
        {
            Console.WriteLine ("No message");
        }
    }
    else
    {
        Console.WriteLine ("The number must be greater than zero");
    }
}
else
{
```

## Type MS · Write a short number program from a list of requirements

### Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 10 of 13 · find part b in this mark scheme extract

# Type MS · Write a short number program from a list of requirements

## Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 11 of 13 · find part b in this mark scheme extract

**Java**

```java
// Q03cFINISHED

import java.util.Scanner;

public class Q03cFINISHED
{
    public static void main (String[] args)
    {
        String aString = "";
        int aNum = 0;
        Scanner myKeyboard = new Scanner (System.in);

        System.out.println ("Enter a number:");

        // Complete the line to take a string input
        aString = myKeyboard.nextLine();

        // Complete the selection statement
        if (aString != "")
        {
            // Convert the string to an integer
            aNum =  Integer.parseInt (aString);
            if (aNum > 0)
            {
                // Complete the test
                if ((aNum <= 20) || (aNum >= 60))
                {
                    System.out.println ("Acceptable");
                }
                // Complete the test
                else if ((aNum >30) && (aNum < 40))
                {
                    System.out.println ("Centre");
                }
                // Complete the test
                else if (aNum == 30)
                {
                    System.out.println ("Perfect");
                }
                else
                {
                    System.out.println ("No message");
                }
            }
            else
            {
                System.out.println ("The number must be greater than zero");
            }
        }
```

## Type MS · Write a short number program from a list of requirements

### Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 12 of 13 · find part b in this mark scheme extract

**Python**

```python
# Q03cFINISHED

# Complete the line to take a string input
aString = input ("Enter a number: ")

# Complete the selection statement
if (aString != ""):

    # Convert the string to an integer
    aNum = int (aString)

    if (aNum > 0):

        # Complete the test
        if (aNum <= 20) or (aNum >= 60):
            print ("Acceptable")
        # Complete the test
        elif (aNum > 30) and (aNum < 40):
            print ("Centre")
        # Complete the test
        elif (aNum == 30):
            print ("Perfect")
        else:
            print ("No message")
    else:
        print ("The number must be greater than zero")
else:
    print ("You must provide a number")
```

## Type MS · Write a short number program from a list of requirements

### Answer 6 of 6 · Question 3 part b · 3 marks

2024 JUN · Paper 02 · extract 13 of 13 · find part b in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|